RAG Assignment: Baseline vs. Re-Ranking

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GitHub repository: `https://github.com/SaiTanooj/ragassign1`

Setup

The document is the Bharatiya Nyaya Sanhita, 2023 (112 pages, about 68,000 words). I split it into 571 chunks of 150 words with a 30-word overlap. Both pipelines use exactly the same chunks.

- **Bi-Encoder:** `all-MiniLM-L6-v2`, with a FAISS inner-product index over normalised vectors (so inner product equals cosine similarity).
- **Cross-Encoder:** `cross-encoder/ms-marco-MiniLM-L-6-v2`.
- **LLM:** Gemini 2.5 Flash (Queries 1 and 2) and `gemini-3.8-flash` (Query 4, run after 2.5 Flash was retired for new users), with the same prompt for both pipelines ("answer using ONLY the context; otherwise say Not found").
- **Pipeline 1:** Top 10 by cosine similarity, then the first 3 go to the LLM (Answer 1).
- **Pipeline 2:** the same Top 10 are re-scored by the Cross-Encoder, and the new Top 3 go to the LLM (Answer 2).

1 Architecture Comparison

A **Bi-Encoder** runs the query and each chunk through the transformer *separately*, and each is pooled into one fixed-size vector. Relevance is then

$$\text{score}(q, d) = \cos(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \, \|\mathbf{d}\|}.$$

Because chunk vectors do not depend on the query, they are computed once and stored. The drawback is that the query tokens and chunk tokens never interact, and a whole passage is squeezed into a single vector, so small but important details such as an "except when…" clause or a negation can get blurred.

A **Cross-Encoder** takes the query and the chunk *together* as one input (`[CLS] query [SEP] chunk [SEP]`). Self-attention runs across all tokens of both, so every query word can attend directly to every chunk word. The output is a single relevance score for that pair. This is more precise, but nothing can be precomputed: every (query, chunk) pair needs its own forward pass.

2 Qualitative and Rank Shift Analysis

2.1 Rank changes

Table 1 shows the rank of the chunk containing the answer (found by matching the exact answer phrase) before and after re-ranking. Only the Top 3 reach the LLM.

Query	Failure type	Baseline	Re-ranked	Result
Is killing always murder?	Exception	#7	#1	Fixed
Is it murder if the victim agreed to die?	Exception	#4	#1	Fixed
Is there any right of self-defence against a public officer doing his duty?	Negation / restriction	#9	#1	Fixed
Can someone who was drunk when committing a crime escape punishment?	Lexical distractor	#1	#8	Hurt

Table 1: Rank of the answer-bearing chunk, baseline vs. Cross-Encoder re-ranking.

Query 1 ("Is killing always murder?"). The answer is Exception 1 (culpable homicide is not murder under grave and sudden provocation). Chunk 190 holds this clause and moved from rank #7 to #1. Chunk 189 ends in the middle of the sentence ("...whilst deprived of the power"), so it was in the baseline Top 3 but did not contain the word "provocation"; the rest of the sentence sits in chunk 190. The Bi-Encoder ranked the general murder/culpable-homicide chunks higher because they match the query topic better.

Query 2 ("Is it murder if the victim agreed to die?"). The answer is Exception 5 (consent of a person above eighteen). Chunk 196 moved from #4 to #1. None of the baseline Top 3 chunks (185, 197, 187) contain the phrase "own consent". The Cross-Encoder also pulled chunk 189 up from #10 to #2.

Query 3 (self-defence vs. public officer). The relevant restriction on the right of private defence moved from #9 to #1. This is a limit on a general right, which is the kind of constraint the Bi-Encoder tends to miss.

2.2 Queries and generated outputs

The outputs below are copied from the notebook (identical prompt in both pipelines; see Setup for the model used). The chunk IDs show which chunks were sent to the LLM.

Query 1: "Is killing always murder?" Baseline Top 3 chunks: 197, 189, 185. Re-ranked Top 3 chunks: 190, 189, 197.

Answer 1 – Baseline (Top 3 by cosine similarity)

No, killing is not always murder.

The context states:

- "A strikes Z, who is in a sound state of health, with a stick, not intending to cause Z's death. Z dies in consequence of the blow. A is not guilty of murder, if he did not intend to cause death, or such bodily injury as in the ordinary course of nature would cause death." [Chunk 189]
- "A, intending to cause a slight injury to Z, strikes him with a stick, Z dies. A has committed the offence of culpable homicide, but not murder." [Chunk 185]
- "Exception 1.—Culpable homicide is not murder if the offender, whilst deprived of the power" [Chunk 189]

Answer 2 – Re-ranked (Top 3 by Cross-Encoder)

No, killing is not always murder.

[Chunk 189] states that if A intends to cause bodily injury but not death, and the injury is not such as would ordinarily cause death, then even if Z dies in consequence, A is "not guilty of murder".

[Chunk 189] and [Chunk 190] also introduce "Exception 1", which states: "Culpable homicide is not murder if the offender, whilst deprived of the power of self-control by grave and sudden provocation, causes the death of the person who gave the provocation or causes the death of any other person by mistake or accident".

Query 2: "Is it murder if the victim agreed to die?" Baseline Top 3 chunks: 185, 197, 187. Re-ranked Top 3 chunks: 196, 189, 197.

Answer 1 – Baseline (Top 3 by cosine similarity)

Not found in the context.

Answer 2 – Re-ranked (Top 3 by Cross-Encoder)

Culpable homicide is not murder when the person whose death is caused, being above the age of eighteen years, suffers death or takes the risk of death with his own consent.

Query 3: "Can someone who was drunk when committing a crime escape punishment?" Baseline Top 3 chunks: 82, 277, 73. Re-ranked Top 3 chunks: 388, 398, 405. This is the case where re-ranking made the answer worse.

Answer 1 – Baseline (Top 3 by cosine similarity)

Yes, a person who is intoxicated can avoid liability/punishment if, by reason of intoxication, they are incapable of knowing the nature of the act, or that they are doing what is wrong or contrary to law, provided that the intoxicating substance was administered to them without their knowledge or against their will.

Additionally, for offences requiring particular intent or knowledge, they are liable to be dealt with as if they had the same knowledge as if they were sober, unless the intoxicant was administered to them without their knowledge or against their will.

Answer 2 – Re-ranked (Top 3 by Cross-Encoder)

Not found in the context.

2.3 Comparison of Answer 1 and Answer 2

- **Query 1:** Answer 1 correctly said "No", but it only quoted the cut-off fragment of Exception 1 and never mentioned the actual condition (grave and sudden provocation) or its provisos, so an essential clause was omitted. Answer 2 quoted the complete Exception 1 because chunk 190 was in its context.
- **Query 2:** Answer 1 was "Not found in the context" even though the answer exists in the document, because chunk 196 was at rank #4. Answer 2 correctly stated that it is not murder when a person above eighteen suffers death or takes the risk of death with his own consent.
- **Query 3 (reverse case):** here Answer 1 was correct (intoxication is a defence only if the person was incapable of knowing the nature of the act, and the intoxicant was given without his knowledge or against his will), but Answer 2 was "Not found in the context", because re-ranking replaced the correct chunk with three chunks about escaping custody.

In none of the cases did either pipeline invent content; the "use ONLY the context" instruction made the pipeline with the wrong chunks omit information or refuse. The errors come from which

chunks reach the LLM, not from the LLM itself.

2.4 Cases where re-ranking did not help

Re-ranking is not an automatic improvement. Table 2 lists the queries from my rank check where it did not help, and they fall into three groups.

Query	Baseline	Re-ranked	Outcome
Can someone who was drunk when committing a crime escape punishment?	#1	#8	Hurt
What is the punishment for stealing a child?	not in Top 10 (true #16)	–	Missed by stage 1
Is there a limit on how much force I can use to defend myself?	not in Top 10 (true #33)	–	Missed by stage 1
Does the right of self-defence allow causing more harm than necessary?	not in Top 10 (true #12)	–	Missed by stage 1
Can I hit a police officer who is arresting me?	not in Top 100	–	Missed by stage 1
Is it a crime to keep money I found on the road?	#7	#7	Both fail
What is the punishment for taking a child away from the parents without their permission?	#4	#6	Both fail

Table 2: Queries where Cross-Encoder re-ranking did not put the answer chunk in the Top 3.

1. Re-ranking made things worse (drunkenness query). The baseline put the correct chunk at #1 and the Cross-Encoder moved it to #8, which changed a correct LLM answer (Answer 1) into "Not found in the context" (Answer 2). I ran several checks to find out why:

- **What replaced it.** The new Top 3 (chunks 388, 398, 405) are all about escaping from custody or resisting apprehension (harbouring an escaped offender, a public servant letting a prisoner escape, and so on). None of them mention intoxication, so these were not alternative valid answers. They share the word "escape" with my query ("escape punishment"), but in the query it means *avoid liability*, while in these chunks it means *run away from custody*.
- **Size of the gap.** The answer chunk scored -5.97 against -1.10 for the new #1, so this was not a narrow miss. In contrast, the Bi-Encoder gave the answer chunk a clear lead (cosine 0.557 vs. 0.483 for the next chunk), so it matched "drunk" to "intoxication" correctly.

Taken together, the Cross-Encoder appears to rely heavily on literal word overlap between query and passage. The document says "intoxication" where I wrote "drunk", so it had no matching word to hold on to, and it followed the word "escape" instead. This is the keyword-trap situation described in the assignment, except that here the Bi-Encoder handled the synonym and the Cross-Encoder was the one that fell for the distractor. Both Cross-Encoders I tried come from the same MS MARCO family, so I cannot say whether a re-ranker trained differently would behave better; I did not test that.

2. The answer was never in the Top 10. For four queries the answer chunk sat at rank #12, #16, #33 or beyond the Top 100 in the baseline, so it never reached the Cross-Encoder at all. Re-ranking only reorders the 10 candidates it receives, so the final result can never be better than the recall of the first stage. Here the cause seems to be a vocabulary gap between everyday language ("hit a police officer", "stealing a child") and the legal wording of the Sanhita ("colour

of his office", "kidnaps... from lawful guardianship"). Rewording the same questions closer to the legal text helped the first stage: "Is there any right of self-defence against a public officer doing his duty?" ranked the answer at #9 (and re-ranking then moved it to #1), while the everyday version, "Can I hit a police officer...", never appeared in the Top 100. A larger candidate pool (for example Top 50 instead of Top 10) would have included the #33 chunk, but it would also cost more Cross-Encoder time (about 0.06 s per pair from Table 3, so around 3 s for 50 pairs). I did not test whether the Cross-Encoder would then rank it correctly.

3. Both stages failed. For the "money found on the road" and "taking a child away" queries the answer chunk was inside the Top 10 (#7 and #4) but neither the Bi-Encoder nor the Cross-Encoder placed it in the Top 3; the second query even moved from #4 to #6. Here the Cross-Encoder had the right chunk available and still did not promote it.

Overall, re-ranking fixed the three queries in Table 1 where the answer was already somewhere in the Top 10 but buried, and it did not help (or hurt) when the first stage missed the chunk or when the Cross-Encoder itself scored the wrong chunk higher. In my results, the Cross-Encoder helps most when an exception or restriction chunk is retrieved but ranked low.

3 Latency Profiling

Times were measured with `time.perf_counter()` on CPU (PyTorch CPU build). "Baseline" includes embedding the query and the FAISS search over the 571 chunks. "Re-rank" is the Cross-Encoder scoring 10 pairs. LLM generation time is not included.

Query	Baseline (s)	Re-rank (s)	P2 total (s)
Killing query (avg of 5 runs)	0.028	0.577	0.605
Killing query (single run)	0.027	0.569	0.596
Victim consent query	0.029	0.615	0.644
Sudden quarrel query	0.034	0.626	0.660
Policeman on duty query	0.027	0.637	0.664

Table 3: Runtime of Pipeline 1 (baseline retrieval) and the extra time added by Pipeline 2 (re-ranking).

Re-ranking adds roughly 0.6 s per query, which is about 20 times the baseline retrieval time. For this use case the extra delay is small in absolute terms, but it dominates the retrieval cost.

4 Why Not Run the Cross-Encoder on the Whole Collection?

A Cross-Encoder needs a full forward pass for every (query, chunk) pair, and since the query is part of the input, chunk representations cannot be computed in advance. Cost therefore grows linearly with the number of chunks *for every single query.* From Table 3, scoring 10 pairs takes about 0.58 s, i.e. about 0.06 s per pair. For my 571 chunks that would be roughly 33 s per query (my linear extrapolation), and for a collection with millions of chunks it would be completely impractical.

A Bi-Encoder avoids this: all chunk vectors are computed once offline, and at query time only the query is encoded and a fast (possibly approximate) nearest-neighbour search is done, which here took about 0.03 s. The two-stage design uses each model for what it is good at: the Bi-Encoder gives fast, high-recall candidate retrieval, and the Cross-Encoder spends its expensive computation only on the small shortlist to get precise ordering. The limitation, as seen above, is that the final quality is capped by the recall of the first stage.